



Many steps to get to this point

ries: deposition, removal, patterning and modification of electrical properties.

- **Deposition** - This refers to any process which grows, coats or in some other way transfers a material on to a wafer. Some of the commonly used technologies are chemical vapour deposition (CVD), physical vapour deposition (PVD) and molecular beam epitaxy (MBE).
- **Removal** - This refers to any process using which material is removed from the wafer. An example is the etching processes – either dry or wet.
- **Patterning** - This is the step by which the deposited materials are shaped or altered. It's usually called lithography. In conventional lithography, the wafer gets coated with a chemical. After this, a machine termed 'stepper' would focus, align and move a mask. This exposes particular portions of the wafer lesser than the short wavelength light. The exposed regions would then get washed away with a developer solution. Once the etching (or similar process) gets done, the leftover chemical is also removed using a technique called plasma ashing.
- **Modification of electrical properties** - One of the key elements of this step is doping transistor sources and drains. The doping processes would be followed by furnace annealing. In advanced devices, the process would be rapid thermal annealing. Annealing helps the implanted dopants to be activated. These days, the step also includes reducing a material's dielectric constant in low-k insulators. This is done by exposing to ultraviolet light in UV processing (UVP).

Modern chips could contain up to eleven metal levels. To produce this may require more than 300 sequenced processing steps.